

# PANKAJ

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## Career Objective

Highly motivated B.Tech Computer Science Engineering student specialized in Cybersecurity. Deeply passionate about competitive programming, Data Structures & Algorithms (DSA), and systems architecture. Proficient in C++ and Java; seeking a software engineering or security internship to build secure, robust, and high-performance systems.

## Education

- **VIT Bhopal University**
  - B.Tech in Computer Science Engineering (Cybersecurity & Digital Forensics) | Graduation: 2027
  - CGPA: 8.3 / 10

## Technical Skills

- **Languages:** C++ (Advanced - DSA, STL, OOP), Java, Python, SQL
- **Frameworks & Core:** Spring Boot, Spring Core, Spring AOP, Data Structures & Algorithms, OOP
- **Tools & Ecosystem:** Maven, Git, GitHub, Linux, Wireshark, MySQL, VS Code, CMake, PowerShell

## Technical Projects

[Low-Latency C++ String Engine](#) | *Live GitHub Repository*

*Tech: C++, GCC/MinGW, SIMD (AVX2), Git, System Memory Management*

- **SSO Architecture:** Built a custom string engine leveraging Small String Optimization (SSO) to bypass dynamic heap allocations for short strings by rendering them directly on the stack block.
- **Memory & SIMD Efficiency:** Implemented an automated fallback routine for scaling past stack limits alongside zero-copy token views, compiling with `-mavx2` and `-O3` flags to ensure cache locality.

[Spring Boot Core Backend Engine](#) | *Live GitHub Repository*

*Tech: Java 17, Spring Boot 3.x, Spring AOP, Maven, YAML, Git*

- **Framework Architecture:** Designed a core backend utilizing lifecycle hooks (`@PostConstruct/@PreDestroy`), explicit constructor injection, and profile-driven configurations (`@ConditionalOnProperty`).
- **Aspect Interception:** Implemented a runtime execution logging layer using custom annotations (`@LogExecutionTime`) and AOP proxies.

[JVM Bootstrapping & Custom ClassLoading Sandbox](#) | *Live GitHub Repository*

*Tech: Java, Java Reflection API, Bytecode Manipulation, Git, PowerShell*

- **Dynamic Loading:** Engineered a custom `ClassLoader` by overriding native `findClass()` and `defineClass()` mechanics for binary `.class` stream parsing.
- **Runtime Isolation:** Analyzed JVM hierarchies (Bootstrap, Platform, Application) to establish security boundaries, using the Reflection API to run decoupled modules without compile-time dependencies.

## Experience & Strengths

- **Cybersecurity Training Program** | *IBM - Adroit Technologies (2 Months)*
  - Trained in enterprise security operations, data privacy paradigms, and packet-level threat hunting via Wireshark.
- **Algorithmic Problem Solving:**
  - Solved 200+ complex DSA problems across competitive programming platforms using C++ and Java, focusing on algorithmic optimizations and space/time minimization.

## Certifications

- Programming in Java (Professional Certification)
- Microsoft SC-900 (Security Fundamentals)
- Blockchain and its Applications (NPTEL)
- IBM Certified Specialist (CECSA1IN Cybersecurity)
- Google IT Support Professional | Python Programming Specialist